# Assignment 01

For this assignment you will be writing 3 programs which should be saved independently as their own ".py" file. The filename you should use for each program is outlined in the sections below. When you're finished you should submit your programs to the Assignment #1 category inside of Brightspace.

You may work by yourself or with a partner. If you choose to work with someone else please be sure to give them credit in your source code and on Brightspace when you submit your work. Each member of a team must submit their own copy of the assignment to get credit for the assignment.

Note that you must include comments in each program and each program must be properly named as explained below. The naming convention is the same for each program - LastNameFirstName_assignX_problemY.py (where ‘X’ refers to the assignment number and ‘Y’ refers to the problem number; for example,"SteigmanAmanda_assign1_problem1.py"). **You will receive 5 points on this first assignment for including reasonable comments in each script and 5 points for naming the files correctly.**

## Problem #1: Variables & Print Statements

Given the following variables:

```
price1 = 10

price2 = 26

price3 = 18

first_name = "Carlos"

last_name = "Alcaraz"

discount = .2
```

Write a program that generates the output below. Ensure that your output matches the sample output exactly, including spaces. You MUST use the variables given (e.g. you can't use the numeric literal 26 in your program - you must refer to the variable `price2`. Note that there is no user input required in this program.

```
Shopping cart for: Carlos Alcaraz

Item prices: 10, 26, and 18

Average Price: 18.0

Total Price with Discount: 43.2
```

Comment your source code and describe your code to someone who may be viewing it for the first time. Also include your name, the date, your class section and the name of your program at the top of your file.

This program should be named as follows:

LastNameFirstName_assign1_problem1.py (for example,"SteigmanAmanda_assign1_problem1.py")

**Question 1 Points (20)**

- Given input variables are used; i.e. no hard coding of names or numbers **(5 points)**
    - Note that you may include other variables in addition to this
- Prior to printing, the individual prices are integers and the discount and total price with discount are floats **(5 points)**
- The output EXACTLY matches the given output in the pdf file; i.e. proper spacing and punctuation **(10 points)**

# Problem #2: Using the print function

Prompt the user to enter 3 sports. Store each sport in a separate variable. Then display the sports back to the user in every possible order using the output below as a guide. Note the formatting in the output below - make your program match the formatting exactly. Items that are underlined indicate that the user has input those values (you don't need to replicate the underlines in your program)

Sample input/output:

```
Please enter sport #1: Basketball

Please enter sport #2: Soccer

Please enter sport #3: Tennis


Here are your sports in every possible order:

-------------------------------------------


1. Basketball, Soccer, Tennis


2. ***Basketball*** ***Tennis*** ***Soccer***


3. Soccer:Basketball:Tennis


4. Soccer

Tennis

Basketball
```

```
5. Tennis?
   Soccer!
   Basketball?!


6. --Tennis
   -----Basketball
   -------Soccer
```

Comment your source code and describe your code to someone who may be viewing it for the first time. Also include your name, the date, your class section and the name of your program at the top of your file.

Note: your program should not print out the sports Basketball, Soccer, Tennis -- it should print out whatever sports the user decides to enter. For example, if I enter Football, Baseball, and Swimming, the output would change to reflect these sports.

Hint: Make sure that IDLE is set up to use a fixed-width font (Courier is the font that I use in class) and that your font size is set to 14pt. On a Mac you can click on IDLE->Preferences to adjust your font size; On a PC you can click on Options->Configure IDLE

This program should be named as follows:

LastNameFirstName_assign1_problem2.py (for example,"SteigmanAmanda_assign1_problem2.py")

**Problem 2 Points** (30)

- The input function is properly used to ask the user for 3 sports, each of which is saved as a variable **(5 points)**
- Each of the 6 possible orders of sports is printed **(6 points)**
- The output of each of the 6 orderings EXACTLY matches the given output in the pdf file; i.e. proper spacing and punctuation **(18 points; 3 points each)**
- Header is printed out **(1 point)**

# Problem #3: Math Expressions

You have recently been hired by a local surveyor to create a program that simplifies the conversion of length measurements. To help you get started, the surveyor has provided you with a list of common conversions:

- 1 mm = 0.0394 in.
- 1 link = 7.92 in.
- 1 ft. = 12 in.
- 1 yd. = 3 ft.
- 1 rod = 25 links
- 1 chain = 4 rods
- 1 mile = 1760 yds.

Using the conversions above, design a program that converts 750 millimeters into US Length units. Here's a sample running of your program - ensure that your output matches the output below (i.e. your items should line up in columns just like the output displayed below). Do not worry about decimal point precision for this program (no numeric formatting required)

**No user input is required for this program.**

Hint: Make sure that IDLE is set up to use a fixed-width font (Courier is the font that I use in class) and that your font size is set to 14pt. On a Mac you can click on IDLE->Preferences to adjust your font size; On a PC you can click on Options->Configure IDLE

```
------------------------------------------------------------
                          mm to US Length Units
------------------------------------------------------------
             mm:                        750.0
             in:                        29.5276
             links:                     3.731348458
             ft:                        2.46063333
             yds:                       0.82021111
             rods:                      0.14912929
             chains:      0.037282323
             mi:                        0.000466029
------------------------------------------------------------
```

Comment your source code and describe your code to someone who may be viewing it for the first time. Also include your name, the date, your class section and the name of your program at the top of your file.

This program should be named as follows:

LastNameFirstName_assign1_problem3.py (for example,"SteigmanAmanda_assign1_problem3.py")

**Problem 3 Points (40)**

- All conversion rules are expressed in variables **(10 points)**
- The output format is generally followed; i.e. one line per conversion; header and footer included **(10 points)**
    - Note that formatting is not super important for this problem, so if things aren't properly lined up and numbers include arbitrary numbers of decimal points, that's okay
- Each number in output matches the given output in the pdf file within the precision of +/- .1 **(20 points; 2.5 points each)**